Reverse-Engineering and Analysis of a Proprietary Health Score from Raw Biometric Inputs

Comparative Evaluation of Modelling Approaches and
Synthetic Data Augmentation

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Chapter 1

Summary

Commercial body scanning platforms such as Anovator produce proprietary composite health scores whose computational basis is undisclosed. This study treats the Anovator bodyAge score as a reverse-engineering target rather than a validated clinical measure, asking whether machine learning regression models can learn to reproduce it from the raw biometric inputs the platform records. The target variable is the age gap, defined as bodyAge minus chronological age. Data comprise 158 scan records from 53 unique clients collected at Zdravoletie, a health and rehabilitation centre in Bulgaria.

Three experiments were conducted using 30 repeated GroupShuffleSplit cross-validation iterations grouped by client identity. Experiment 1 compared five regressors (RandomForest, Ridge, SVR, GradientBoosting, and HistGradientBoosting) on the full 88-feature set. HistGradientBoosting achieved the lowest mean MAE of 2.43 years (95% CI [2.09, 2.76]), but none of the ten pairwise Wilcoxon signed-rank tests reached the Bonferroni-corrected significance threshold of 0.005; all model comparisons are inconclusive at this sample size. Experiment 2 tested three training regimes (real data only, real plus 1000 GMM-generated synthetic records, and synthetic only), finding that synthetic augmentation significantly reduced MAE for four of five models (adjusted $p < 0.003$) but degraded Ridge under the synthetic-only condition from 2.48 to 3.23 years, a consequence of pseudo-label variance compression that does not affect tree-based methods. Experiment 3 evaluated four cumulative feature groups using HistGradientBoosting, finding that the 63-feature anthropometric and metabolic base group alone achieved the lowest mean MAE of 2.39 years, with no statistically significant improvement from adding bioimpedance, postural risk, or joint angle features, suggesting these modalities carry negligible independent weight in the Anovator formula at this data scale.

The primary limitation is dataset size: 53 individuals is insufficient for high-power pairwise significance testing after Bonferroni correction, and most comparisons must be characterised as inconclusive rather than resolved.

The thesis is structured as follows. Chapter 2 establishes the research context and practical motivation. Chapter 3 states the research questions and describes the Streamlit deliverable. Chapter 4 reviews the relevant literature on biological age prediction, synthetic data generation,

and model interpretability. Chapter 5 documents the full experimental methodology. Chapter 6 reports results and sub-question conclusions. Chapter 7 discusses the findings, their implications, and recommendations for Zdravoletie.

Chapter 2

Reason for and Relevance of the Research

2.1 Research Context

The past decade has seen a proliferation of commercial health assessment platforms that transform raw biometric measurements into composite health scores intended for use in wellness, rehabilitation, and preventive health contexts. These platforms, combining bioelectrical impedance analysis, postural imaging, anthropometric measurement, and proprietary scoring algorithms, are deployed in gyms, clinics, and corporate health programmes, where their outputs are presented to clients and practitioners as meaningful summaries of physiological condition. The Anovator body scanning system, used in this study, is representative of this category: it produces a score called bodyAge, expressed as an estimated biological age in years, derived from a combination of body composition, postural, impedance, and biometric inputs through an undisclosed computational formula.

The interpretability gap created by this architecture is practically significant. A client whose Anovator scan returns a bodyAge five years above their chronological age has been told something consequential about their health, but neither the client nor the attending practitioner can inspect the mechanism that produced this number. Which measurements contributed most to the result? Is the score driven primarily by body fat distribution, postural risk, or impedance characteristics? How would a change in a specific biometric (achieved through a targeted intervention) translate into a change in the score? Without access to the proprietary formula, none of these questions can be answered by examining the score alone. This is not unique to Anovator: the same interpretability gap exists across the broader class of commercial health scoring devices, and it limits the evidential value of their outputs for clinical reasoning.

2.2 Research Framing

Addressing the interpretability gap requires a reverse-engineering approach. Rather than predicting biological age in the clinical sense (where ground truth is established against mortality or morbidity endpoints, as in the biological age literature reviewed in Chapter 4), the study treats the Anovator age gap - the signed difference between bodyAge and chronological age, as defined in Section 5.2.2, as the prediction target and asks whether machine learning regression models can learn to reproduce it from the raw biometric inputs that the Anovator system records. Three research sub-questions that structure the empirical programme are stated in Chapter 3.

2.3 Study Design and Scope

The empirical programme comprises three experiments conducted on a dataset of 158 Anovator scan records drawn from 53 unique clients at Zdravoletie, a health and rehabilitation centre in Burgas, Bulgaria. Data were collected between 2022 and 2024. All experiments — each being an empirical comparison of models, training regimes, or feature groups conducted under identical controlled conditions — use 30 repeated GroupShuffleSplit cross-validation iterations grouped by client identity to prevent person-level data leakage, and report predictive performance as mean absolute error (MAE) with 95% confidence intervals. Statistical significance between experimental conditions (the models compared in Experiment 1, the training regimes in Experiment 2, and the feature groups in Experiment 3) is assessed using the Wilcoxon signed-rank test with Bonferroni correction for multiple comparisons.

The scope of the project is deliberately constrained. It covers exactly the three experiments described above. What lies outside the scope is equally important to state clearly. This project does not attempt to validate the Anovator bodyAge score as a clinical measure of biological ageing; the absence of an independently validated ground truth makes such validation impossible within this dataset, and the research question does not require it. The study does not investigate alternative synthetic data generation methods beyond the Gaussian Mixture Model. CTGAN, vine copula, and variational autoencoder approaches are noted in Chapter 4 but excluded from empirical comparison on the grounds that the 158-record training corpus falls below the regime in which these methods train reliably without extensive hyperparameter search. The study does not explore time-based train-test splits, as the dataset contains insufficient repeated measurements per individual across time to make generalisation to future scans from the same individual the primary evaluation criterion. These exclusions are not omissions; they are boundaries that keep the empirical claims within what the data can support — for example, which algorithm best approximates the score, or whether synthetic augmentation improves prediction quality.

2.4 Practical Significance

Understanding the informational structure of a proprietary health score has concrete value even in the absence of access to the formula itself. If reverse-engineering experiments reveal that the anthropometric and metabolic feature group alone accounts for essentially all predictable variation in the Anovator bodyAge score, as Experiment 3 suggests, then practitioners can infer that the specialised measurement modalities contribute negligible marginal information to the score and focus their interpretive attention accordingly. More broadly, the methodology developed here (reverse-engineering a proprietary score through grouped cross-validation, statistical significance testing under multiple comparisons, and feature group ablation) is transferable to any context in which a commercial health device produces a black-box output that practitioners wish to understand without access to the underlying formula.

Chapter 3

Research Questions and Product Description

3.1 Research Questions

The central research question is: can machine learning regression models learn to reproduce the Anovator bodyAge score from the raw biometric inputs that the Anovator scanning platform records? The target is the age gap, defined as bodyAge minus chronological age, across 158 scan records from 53 unique clients collected at Zdravoletie.

The sub-questions presented here were reformulated after the midterm review, following the project reframing described in Section 1.2 of the competency portfolio. The original sub-questions reflected an early exploratory framing that predated the identification of the circular benchmarking error and the shift to a reverse-engineering research objective — they addressed scanner inconsistencies, body ratio predictors, model interpretability, and population simulation in practitioner-facing terms. The reformulated sub-questions below are directly answerable by the three experiments, produce verifiable statistical conclusions, and are fully aligned with the corrected reverse-engineering framing.

Three sub-questions structure the empirical programme.

Sub-question 1. Among five regression algorithms spanning linear, kernel, and ensemble methods (RandomForest, Ridge, SVR, GradientBoosting, HistGradientBoosting), which best approximates the Anovator age gap from the full 88-feature biometric set, and can any observed performance differences be established as statistically significant at the available sample size?

Sub-question 2. Does augmenting the 158-record training set with synthetic records generated by a Gaussian Mixture Model and pseudo-labelled by the production model improve approximation quality on real held-out data, and if so, for which regression model classes?

Sub-question 3. Across four cumulative groups of biometric features — where cumulative means each group is evaluated by adding its features to all preceding groups, so that the contribution of each new group is measured against the best model achievable without it — (anthropometric and metabolic, bioimpedance, postural risk, and joint angles), which groups carry independent predictive information relative to the Anovator score, and which are redundant with what the base group already provides?

3.2 Product Description

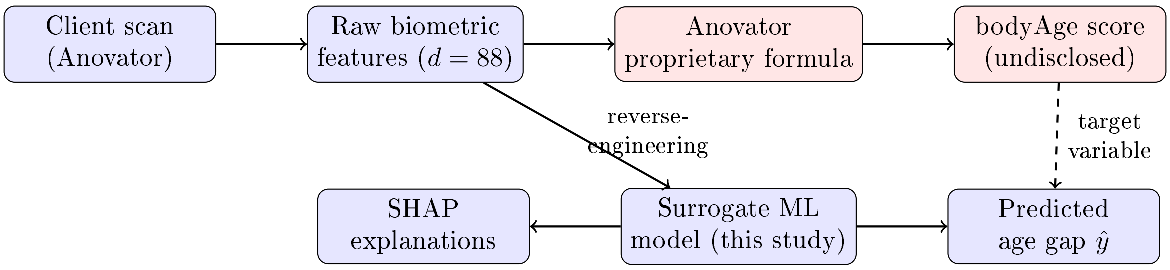

Figure 3.1: Overview of the reverse-engineering chain. The Anovator platform transforms raw biometric features into a proprietary bodyAge score through an undisclosed formula (red boxes). This study trains a surrogate ML model on the same raw features to reproduce the age gap output and generate SHAP explanations (blue boxes). The dashed arrow indicates that the bodyAge score serves as the target variable for the surrogate model.

The research deliverable beyond the three experiments is a multi-page Streamlit web application built for operational use at Zdravoletie. The application accepts an Anovator client export file via a drag-and-drop upload interface and provides three analytical views. The Population View presents aggregate statistics and visualisations across all uploaded scan records, allowing practitioners to inspect the distribution of bodyAge scores and age gaps in the client cohort. The Individual Analysis view generates a SHAP bar chart (based on the Shapley Value) that decomposes the model's predicted age gap into per-feature contributions, giving the attending practitioner a ranked explanation of which biometric measurements drove the score for a specific client. Scan reports can be exported to a formatted PDF for inclusion in client records. The interface is bilingual, with all labels and explanatory text available in both Bulgarian and English to accommodate clinic staff and clients.

The application is publicly accessible at `https://zdravoletie-app-r59gvphlkih7549cgipbgt.streamlit.app/`.

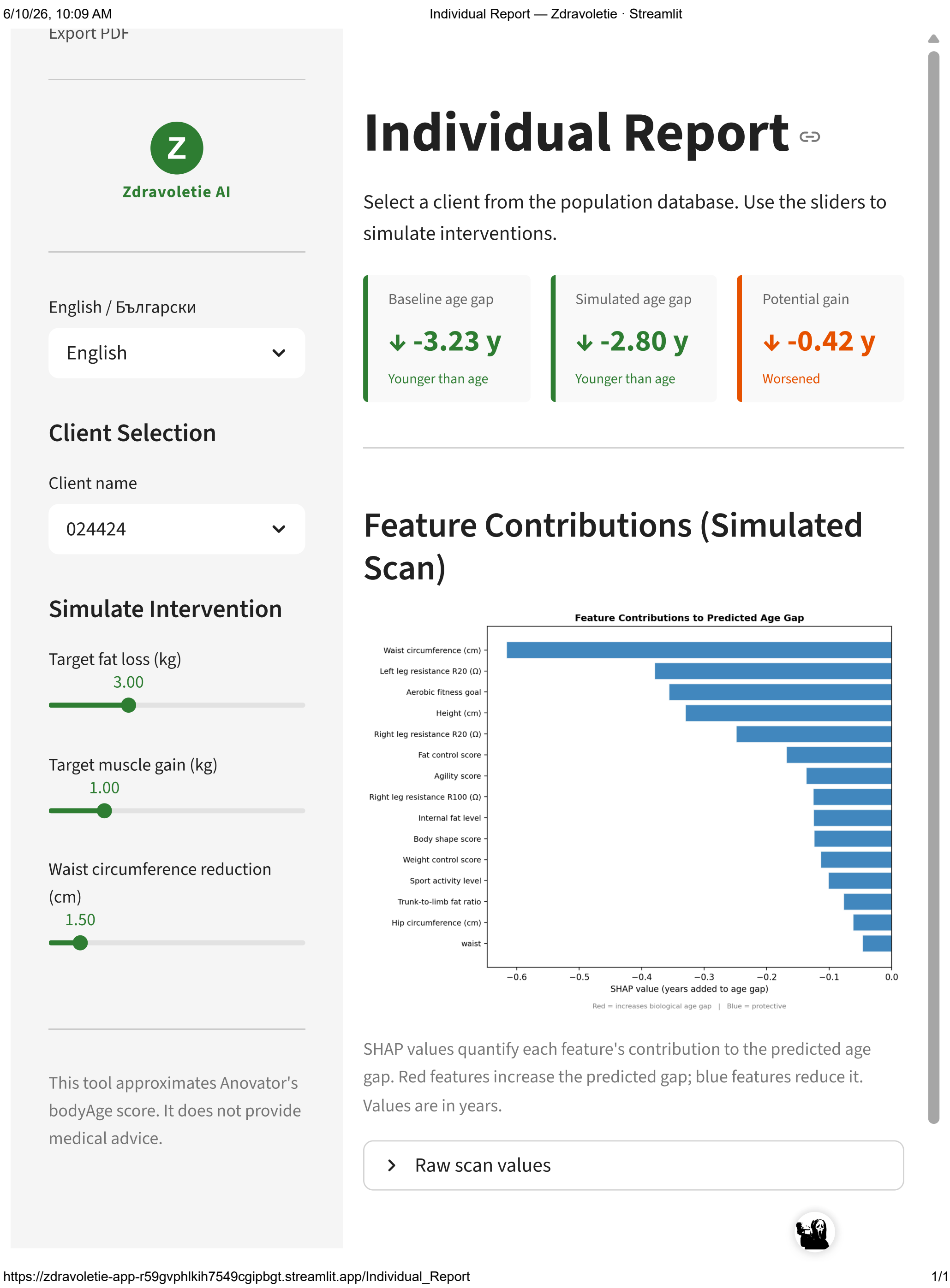

Figure 3.2: Individual Report page of the Zdravoletie Streamlit dashboard. The three colour-coded metric cards show the baseline age gap, simulated age gap after intervention, and potential gain. The SHAP bar chart below displays the top feature contributions to the predicted age gap for the selected client, with human-readable feature labels. Negative values (blue) indicate features that reduce the predicted age gap; positive values (red) indicate features that increase it.

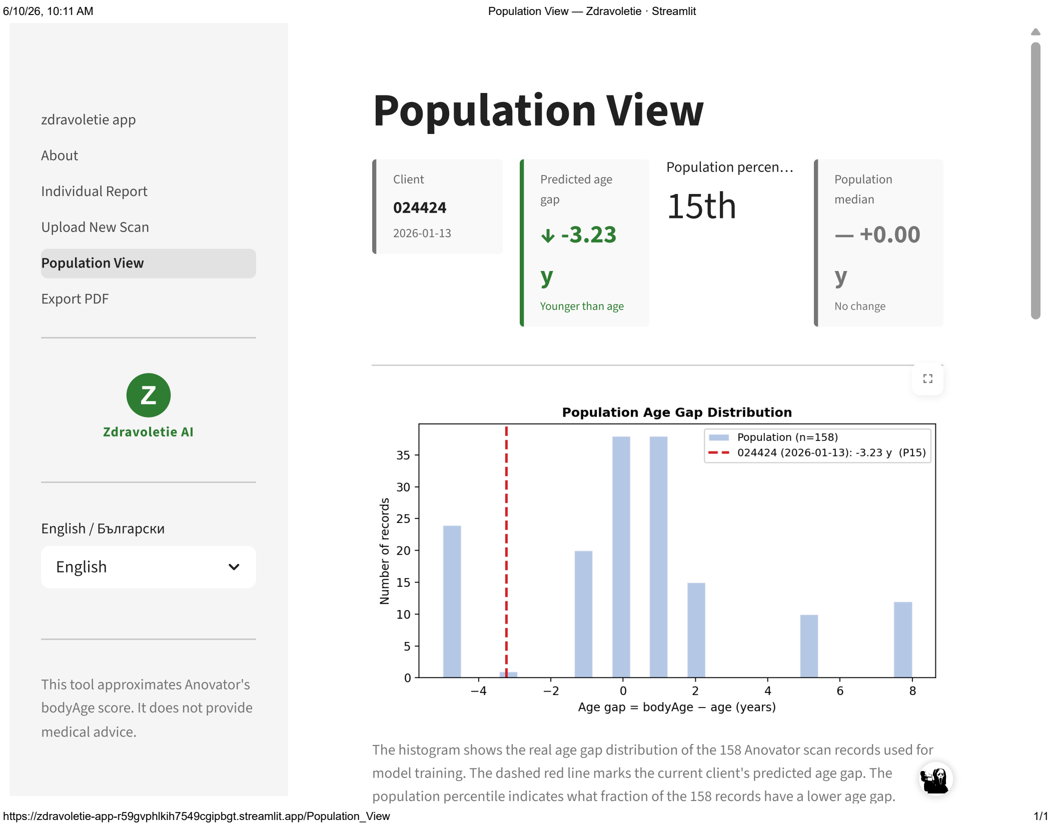

Figure 3.3: Population View page showing the distribution of predicted age gaps across all 158 records in the dataset. The red dashed vertical line marks the currently selected client's predicted age gap, and the legend shows the client's percentile rank within the population. This view allows practitioners to contextualise an individual client's result relative to the full Zdravoletie cohort.

Chapter 4

Theoretical Framework

4.1 Biological Age Prediction

The quantification of biological age (an individual's physiological age as distinct from their chronological age) has been an active area of research in gerontology and clinical epidemiology for several decades. Early attempts used simple composite indices of physical performance or blood chemistry, but the foundational theoretical treatment of the problem was provided by Klemera and Doubal (2006), who formalised biological age estimation as a regression problem in which the true biological age is treated as a latent variable estimated from a set of biomarkers. Their method fits a collection of simple linear regressions, one per biomarker (such as blood pressure, grip strength, lung capacity, or serum albumin level), modelling each biomarker as a function of chronological age, and then combines these fits into a composite biological age estimate that is optimal in the sense of minimising the mean squared distance between the estimated biological age and the biomarker values. Klemera and Doubal (2006) demonstrated that their method produces lower estimation error than earlier composite approaches and explicitly accounts for the fact that biomarkers have different slopes and residual variances with respect to chronological age.

The field advanced substantially with the development of epigenetic clocks, which exploit the observation that DNA methylation — a biochemical process in which methyl groups are added to specific locations on the DNA strand, altering gene expression without changing the underlying sequence — at specific CpG sites (cytosine-guanine dinucleotide positions in the genome where methylation patterns are particularly stable and age-sensitive) changes in a highly age-predictable manner across tissue types. Horvath (2013) demonstrated that a penalised regression model trained on methylation data from 51 tissue types could predict chronological age with a median absolute error of approximately 3.6 years and that systematic deviations from the predicted age (termed age acceleration) were associated with disease status and mortality risk. Building on this foundation, Levine et al. (2018) introduced PhenoAge, a composite biological age measure derived from nine clinical blood biomarkers combined with chronological age through a Gompertz proportional hazards model trained on mortality data from the National Health and Nutrition Examination Survey. The defining feature of PhenoAge is that it is calibrated against a mortality endpoint: a PhenoAge that exceeds chronological age carries a validated association with elevated all-cause mortality risk. Lu et al. (2019) extended this line of work with GrimAge, an epigenetic clock trained on plasma protein levels and smoking pack-years via DNA methylation proxies (methylation patterns used as indirect predictors of plasma protein levels, since direct protein measurement was not available in the training data), optimised to predict time-to-death. GrimAge demonstrated substantially stronger associations with lifespan and health outcomes than earlier methylation clocks.

The present project differs from this literature in both the ground truth available and the measurement modalities used. The ground truth is not a clinical endpoint but the proprietary bodyAge score computed by the Anovator scanning platform. The research question is accordingly not whether a model can predict how quickly an individual will age in a clinical sense, but whether a model can learn to reproduce the Anovator formula's output from its inputs: a reverse-engineering problem. The distinction between reverse-engineering and clinical prediction is important. Reverse-engineering asks whether a known output (the Anovator bodyAge score) can be reproduced from known inputs (the raw biometric measurements the platform records); it requires no claim about the clinical validity of the target variable. Clinical biological age prediction, by contrast, requires a validated ground truth established against mortality or morbidity endpoints. Because the Anovator bodyAge score has not been validated against any such endpoint, treating it as a clinical prediction target would be methodologically unjustifiable. The reverse-engineering framing is therefore not merely a presentation choice; it defines the scope of what claims this study is and is not permitted to make.

4.2 Synthetic Data Generation in Small Medical Datasets

The scarcity of labelled data is a recurring constraint in medical machine learning, where data collection is costly, privacy regulations limit sharing, and cohort sizes at individual institutions are often in the hundreds rather than the thousands. The dataset used in this study comprises 158 scan records with 88 features per record; a full description of columns, data types, and measurement levels is provided in Section 5.2. The features span continuous biometric measurements (body composition, impedance, joint angles), ordinal health goal scores, and binary demographic indicators. The target variable `age_gap` is a continuous integer-valued outcome with mean 0.45 years and standard deviation 3.30 years across the 158 records. At this scale, statistical power is limited and model generalisation is constrained by the narrow demographic range of 53 unique individuals at a single centre.

Although the present study addresses a regression problem rather than a classification one, the synthetic data literature originates largely in the context of class imbalance — the condition in which one outcome category is substantially underrepresented relative to another, for example a dataset in which 90% of records have a near-zero age gap and 10% have elevated values. In such cases, models trained without augmentation learn to predict the majority class and fail on the minority. The earliest and most widely used augmentation approach for this problem is SMOTE (Synthetic Minority Over-sampling Technique), introduced by Chawla et al. (2002), which generates new minority-class instances by interpolating between existing instances in feature space. SMOTE operates in a purely local neighbourhood context and does not model the global joint distribution of features. Modelling the joint distribution matters because biometric features are not independent: waist circumference, body fat percentage, and BMI are strongly correlated, and a synthetic record that samples each feature independently would produce biologically implausible combinations — for example, a record with high muscle mass but also high total body fat percentage and low BMI simultaneously. A generative model that captures the joint distribution preserves these inter-feature dependencies in the synthetic population.

Gaussian Mixture Models (GMMs) fit a weighted sum of K multivariate Gaussian distributions to the observed data. Formally, the GMM models the joint density of the feature vector $\mathbf{x} \in \mathbb{R}^d$ as:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{4.1}$$

where π_k are the mixture weights satisfying $\sum_{k=1}^{K} \pi_k = 1$, $\boldsymbol{\mu}_k \in \mathbb{R}^d$ is the mean vector of component k, and $\boldsymbol{\Sigma}_k \in \mathbb{R}^{d\times d}$ is its full covariance matrix. Parameters are estimated by Expectation-Maximisation on the training records. Once fitted, synthetic samples are generated by drawing from the fitted mixture: a component k is sampled with probability π_k, and a synthetic record is then drawn from $\mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. The full covariance matrices preserve inter-feature correlations in the generated samples, which is essential for maintaining the joint distribution of biometric measurements. GMMs are computationally efficient, produce closed-form likelihood estimates, and require relatively few observations to fit stably, properties that make them well suited to small tabular datasets in the low-hundreds-of-records regime.

Xu et al. (2019) introduced CTGAN, a Conditional Generative Adversarial Network specifically designed for tabular data, addressing multi-modal non-Gaussian distributions and class imbalance through mode-specific normalisation and conditional generation. However, GAN training is sensitive to the ratio of training samples to model parameters, and mode collapse becomes substantially more likely when training sample sizes fall below approximately one thousand records per class.

Copula-based approaches, as systematised in the Synthetic Data Vault framework by Patki et al. (2016), model the joint distribution by separately estimating each variable's marginal distribution and using a copula to capture the dependence structure. For this project, a five-component GMM ($K = 5$) with full covariance matrices was selected as the synthesis method. The choice of $K = 5$ was determined by BIC (Bayesian Information Criterion) model selection over $K \in \{2, 3, 4, 5, 6, 7\}$, which penalises model complexity and favoured five components as the best balance between fit quality and parameter count at the 158-record scale. Full covariance matrices were preferred over diagonal or spherical alternatives because the biometric features in this dataset exhibit substantial inter-feature correlation, and restricting the covariance structure would artificially suppress those relationships in the generated samples.

4.3 Feature Importance and Interpretability in Health Machine Learning

The deployment of machine learning models in health-related contexts introduces an obligation of interpretability that is distinct from the pure predictive performance requirements of general machine learning applications.

The SHapley Additive exPlanations framework, introduced by Lundberg and Lee (2017), provides a theoretically grounded approach to feature attribution. SHAP values are derived from cooperative game theory: the contribution of each feature to a particular prediction is computed as the average marginal contribution of that feature across all possible orderings in which features are added to the model. The game-theoretic foundation gives SHAP values three properties that competing attribution methods do not simultaneously satisfy: local accuracy, consistency, and missingness.

Lundberg et al. (2020) developed TreeSHAP, an algorithm that exploits the structure of tree-based ensemble models to compute exact SHAP values in polynomial time, making SHAP analysis computationally feasible for random forests and gradient boosting models. In a landmark clinical application, Lundberg et al. (2018) used SHAP explanations to identify the features driving a gradient boosting model's predictions of sepsis onset in intensive care patients, demonstrating that SHAP could surface clinically meaningful feature importance patterns that were subsequently validated by domain experts.

In the present project, SHAP analysis serves two purposes, both connected to Sub-question 3: globally, it provides an empirical ranking of feature importance that complements and extends the feature group ablation results of Experiment 3 by attributing predictive weight at the individual feature level rather than the group level; locally, it provides the per-client interpretability output delivered through the Streamlit dashboard. SHAP values explain the model's behaviour, not the underlying physiological process: they characterise the information structure of the Anovator score as recovered by the model, subject to the approximation error and feature encoding limitations described in Chapter 5.

Chapter 5

Research Method

5.1 Thesis Framing and Research Objective

The study does not aim to predict biological age in any general sense. Its specific objective is to reverse-engineer and analyse Anovator's proprietary bodyAge score from raw biometric inputs, and to evaluate three dimensions of that modelling problem: which regression algorithm best approximates the score, whether Gaussian Mixture Model synthetic data augmentation improves approximation quality, and which biometric feature groups carry the most predictive information relative to the score.

5.2 Dataset

5.2.1 Source and Collection

All data originate from Anovator body scan sessions conducted at Zdravoletie, a health and rehabilitation centre in Bulgaria. Records were collected between 2022 and 2024 using the Anovator scanning platform, which combines impedance bioelectrical analysis, postural camera imaging, and manual biometric measurements. The dataset comprises 158 individual scan records drawn from 53 unique clients, with each client potentially appearing in multiple records across different sessions. This structure introduces non-independence of observations: records from the same individual share stable biometric characteristics and are therefore not fully independent draws from the population. For linear regression models, non-independent observations violate the assumption that residuals are independent, which would render inference statistics unreliable. This study addresses the non-independence problem through the GroupShuffleSplit cross-validation protocol described in Section 5.4.1, which ensures all records from a given client are assigned together to either the training fold or the test fold, preventing person-level data leakage. This does not eliminate within-person correlation across sessions but prevents the model from exploiting individual-level consistency as a substitute for population-level generalisation. The dataset was assembled manually from the Anovator web interface; no public data sources were combined with it.

5.2.2 Target Variable

Anovator computes a proprietary score called bodyAge for each scan. The target variable throughout all experiments is the age gap, defined as the signed difference between bodyAge and the client's chronological age at the time of the scan:

$$\text{age\_gap} = \text{bodyAge} - \text{age} \tag{5.1}$$

Both `bodyAge` and `age` are recorded as integers in the Anovator export; `age_gap` is therefore also an integer-valued variable, though it is treated as continuous for the purpose of regression modelling.

A positive age_gap indicates that the Anovator system judged the client's biological condition to be older than their chronological age; a negative value indicates the reverse. Across the 158 records, age_gap has a mean of 0.45 years and a standard deviation of 3.30 years.

5.2.3 Feature Set

The final model feature set comprises 88 variables derived from the raw Anovator outputs, decomposed into four groups used in Experiment 3.

Group 1, Anthropometric and metabolic (63 features): Body composition measurements (weight, height, BMI, body fat percentage, muscle mass, bone mass, protein content, water content, subcutaneous fat, intra-abdominal fat, and segmental decompositions), Anovator-computed health goal recommendations, balance and agility scores, vital signs where available, and demographic features.

A note on multicollinearity: many features within Group 1 share the same biological source or are mathematically derived from one another. For example, BMI is computed from weight and height, which are themselves present as separate features; segmental fat measurements sum to total body fat; and muscle mass and fat percentage are inversely constrained by total body composition. When regressors exhibit collinearity, regression coefficients can become unstable or misleading, and models may overestimate the importance of certain variables (Sullivan et al., 2001). This is particularly relevant for Ridge regression, whose regularisation parameter partially mitigates collinearity by shrinking correlated coefficients toward zero, but does not eliminate the instability. Tree-based models are less sensitive to collinearity because they select features greedily at each split rather than fitting global coefficients. The collinearity structure of Group 1 should be borne in mind when interpreting both the Experiment 1 model comparisons and the SHAP feature importance rankings in Section 7.4.

Group 2, Bioimpedance (10 features): Raw impedance measurements at 20 kHz and 100 kHz for each of the four limbs and the trunk (`r20LeftArm`, `r20RightArm`, `r20LeftLeg`, `r20RightLeg`, `r20Trunk`, `r100LeftArm`, `r100RightArm`, `r100LeftLeg`, `r100RightLeg`, `r100Trunk`).

Group 3, Postural risk (10 features): Camera-based postural risk scores (`frontHeadRisk`, `headRisk`, `humpbackRisk`, `kneeRisk`, `pelvisRisk`, `postureRisk`, `shoulderRisk`, `spineRisk`, `side_hunchback_r` and a derived aggregate (aggregated_postural_index).

Group 4, Joint angles (5 features): front_shoulder_angle, front_left_leg_angle, front_right_leg_an side_left_leg_angle, side_right_leg_angle.

5.3 Feature Engineering

Five derived features are computed from raw columns to capture physiological relationships that individual variables cannot express independently. Body composition ratios reflect the balance between tissue types rather than absolute quantities; the aggregated postural index summarises six correlated postural risk scores into a single composite to reduce redundancy within that feature group. These transformations were motivated by domain knowledge of how the Anovator scoring system is likely to weight relative body composition against absolute measurements. Sex is encoded as a binary integer (0 for female, 1 for male). Three body composition ratios are computed: muscle-to-fat ratio (muscle divided by fat plus 0.1), upper-to-lower body muscle ratio (upperBody divided by lowerBody plus 0.1), and trunk-to-limb fat ratio (fatTrunk divided by the sum of the four limb fat measurements plus 0.1). The additive constant of 0.1 in each denominator prevents division by zero and is applied consistently across all contexts. The aggregated_postural_index is the mean of humpbackRisk, spineRisk, pelvisRisk, postureRisk, kneeRisk, and frontHeadRisk.

5.4 Cross-Validation Protocol

5.4.1 Group-Based Splitting

All three experiments use GroupShuffleSplit from scikit-learn with the grouping variable set to the client name field. This prevents data leakage at the person level: all of a given client's records are assigned together to either the training fold or the test fold. Without this constraint, a model could exploit within-person consistency (learning the stable biometric profile of an individual) rather than generalising across individuals, which would inflate performance estimates.

Each split assigns 80% of the 53 unique individuals to training and 20% to test, yielding approximately 11 to 13 test records per split. The split is non-exhaustive and repeated 30 times. All splits use a fixed random seed of 42.

```python
from sklearn.model_selection import GroupShuffleSplit

cv = GroupShuffleSplit(
    n_splits=30, test_size=0.2, random_state=42
)
for train_idx, test_idx in cv.split(X, y, groups=names):
    X_train, X_test = X[train_idx], X[test_idx]
```

Listing 5.1: GroupShuffleSplit configuration used across all three experiments.

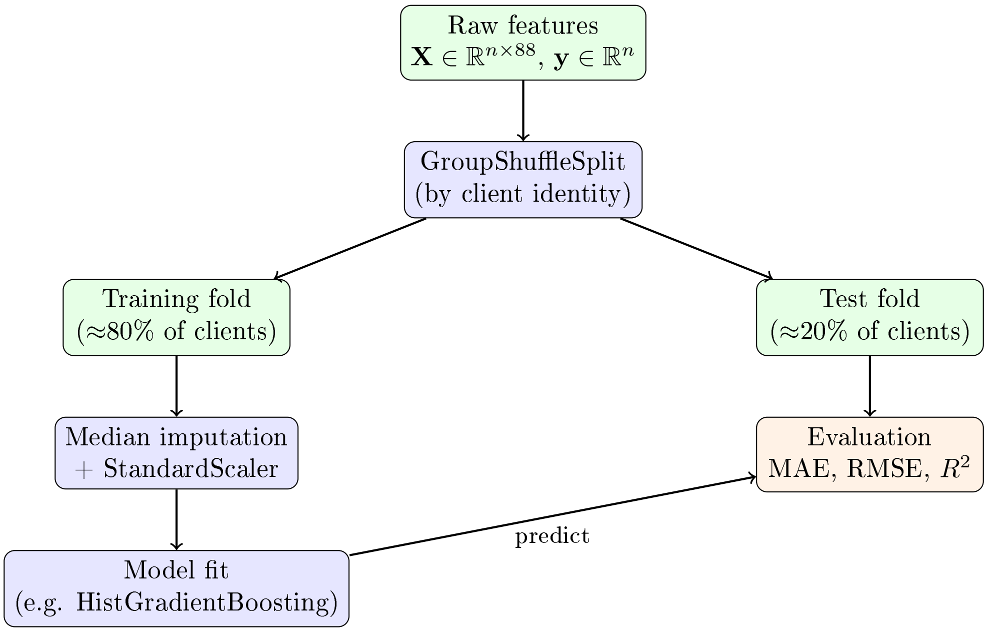

Figure 5.1: Model pipeline schematic for a single cross-validation fold. Raw features $\mathbf{X}$ and target variable $\mathbf{y}$ (`age_gap`) are split by client identity using GroupShuffleSplit. The training fold undergoes median imputation and scaling before model fitting; the same imputation parameters are applied to the test fold. Evaluation metrics are computed on the test fold predictions. This procedure is repeated 30 times with different random splits.

5.4.2 Imputation and Scaling

Missing values are imputed using the median of the training fold in each split, computed fresh from the training fold after the split is drawn. A secondary `fillna(0)` handles the rare case where a feature is entirely missing from the training fold. The test fold is imputed using the same training-fold medians, ensuring that no information from the test set informs the imputation. `StandardScaler` is fit on the training fold after imputation and applied to both folds. Scaling is applied uniformly across all five models for pipeline consistency. Tree-based models (RandomForest, GradientBoosting, HistGradientBoosting) are invariant to monotonic feature transformations and their results are unaffected by this step; Ridge regression and SVR require scaling for correct behaviour, as their optimisation objectives are sensitive to the absolute magnitude of feature values.

```
train_medians = X_train.median()
X_train = X_train.fillna(train_medians).fillna(0)
X_test  = X_test.fillna(train_medians).fillna(0)
scaler  = StandardScaler().fit(X_train)
X_train = scaler.transform(X_train)
X_test  = scaler.transform(X_test)
```

Listing 5.2: Per-fold median imputation preventing data leakage.

5.5 Experiment 1: Model Comparison

Experiment 1 evaluates five regression algorithms spanning linear, kernel, and ensemble families. The mathematical form of each model is described below, with $\mathbf{x} \in \mathbb{R}^d$ denoting the scaled feature vector for a single record and $\hat{y} \in \mathbb{R}$ denoting the predicted age gap.

Ridge Regression. Ridge fits a linear model with ℓ_2 regularisation:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b, \quad \text{where} \quad \mathbf{w}^* = \operatorname*{argmin}_{\mathbf{w}} \sum_{i=1}^{n} (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|_2^2 \tag{5.2}$$

The regularisation coefficient $\lambda = 1.0$ shrinks coefficients toward zero, partially mitigating the instability caused by multicollinear features in Group 1. Ridge serves as the linear baseline in this comparison.

Support Vector Regression (SVR). SVR with a radial basis function (RBF) kernel fits a non-linear model by mapping inputs into a high-dimensional feature space via the kernel function $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$, and finding the flattest function within an ε-insensitive tube around the training targets:

$$\hat{y} = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*)\, k(\mathbf{x}_i, \mathbf{x}) + b \tag{5.3}$$

where $\alpha_i, \alpha_i^* \geq 0$ are dual coefficients. Parameters used: $C = 1.0$, $\varepsilon = 0.1$, RBF kernel with default γ.

Random Forest. Random Forest builds an ensemble of $T = 100$ decision trees, each trained on a bootstrap sample of the training fold with a random subset of features considered at each split. Predictions are averaged across trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(\mathbf{x}) \tag{5.4}$$

where f_t is the t-th tree. Maximum depth is constrained to 7 to limit overfitting on the small training folds.

Gradient Boosting. Gradient Boosting constructs an additive ensemble by fitting each tree f_t to the negative gradient of the loss function with respect to the current ensemble prediction:

$$F_t(\mathbf{x}) = F_{t-1}(\mathbf{x}) + \eta \cdot f_t(\mathbf{x}) \tag{5.5}$$

where $\eta = 0.1$ is the learning rate and f_t is a shallow tree of maximum depth 3. The ensemble uses $T = 100$ trees.

HistGradientBoosting. HistGradientBoosting extends gradient boosting by binning continuous features into a fixed number of histogram bins before tree construction, reducing computational complexity from $\mathcal{O}(nd)$ to $\mathcal{O}(nb)$ per split where $b \ll n$ is the number of bins. The prediction form is identical to Gradient Boosting (Equation 5.5) with up to 100 maximum iterations and native support for missing values.

For each of the 30 splits, fresh model instances are trained on the scaled training fold and evaluated on the scaled test fold. MAE, RMSE, and R^2 are recorded for each model in each split, yielding 30 paired observations per model. MAE is the primary evaluation metric throughout: it is interpretable in the units of the target variable (years), robust to outliers relative to squared-error metrics, and directly meaningful to a clinical audience. RMSE is reported alongside MAE because it penalises large individual errors more heavily, providing a complementary view of prediction reliability. R^2 is reported for completeness only; its interpretation is unreliable at the test fold sizes used here (11–13 records per fold) and it is not used as the basis for any comparative claim.

Summary statistics (mean, standard deviation, and 95% confidence interval) are computed using the t-distribution with 29 degrees of freedom. The 30 fold scores are treated as approximately independent for this purpose: although successive splits share some training records due to the non-exhaustive GroupShuffleSplit sampling, the random reshuffling at each split reduces systematic dependence between fold outcomes, and no stronger independence guarantee is available at this sample size.

5.6 Experiment 2: Synthetic Data Ablation

Experiment 2 investigates whether augmenting the training set with GMM synthetic data improves prediction of age_gap on real held-out data. Three training regimes are evaluated for each of the five models:

- **RealOnly**: trains exclusively on the real records in the training fold.
- **RealPlusSynth**: augments the real training fold with all 1000 synthetic records before training.
- **SynthOnly**: trains exclusively on the 1000 synthetic records and evaluates on the real test fold.

The 1000-record synthetic dataset was generated by fitting a Gaussian Mixture Model with five components and full covariance matrices to the 158 real records, as described in Section 4.2. The GMM learns the joint distribution of all 88 biometric features and generates new records by sampling from that distribution, producing synthetic clients whose feature combinations are statistically consistent with the real population without replicating any individual real record. Pseudo-labels were generated by applying the production model (trained on all 158 real records) to the synthetic feature vectors, constituting a teacher-student knowledge distillation setup. A pseudo-label is the model's predicted `age_gap` value for a synthetic record — for example, a synthetic record whose biometric features resemble a moderately overweight 45-year-old client might receive a pseudo-label of +2.1 years, meaning the production model predicts that the Anovator formula would return a bodyAge 2.1 years above chronological age for that feature combination. The pseudo-labels have a mean of 0.75 years and a standard deviation of 1.91 years, compared to the real label distribution of mean 0.45 years and standard deviation 3.30 years.

```
from sklearn.mixture import GaussianMixture

gmm = GaussianMixture(
    n_components=5, covariance_type='full', random_state=42
)
gmm.fit(X_real)
X_synthetic, _ = gmm.sample(n_samples=1000)
```

Listing 5.3: GMM initialisation and synthetic population generation.

5.7 Experiment 3: Feature Group Ablation

Experiment 3 evaluates the marginal contribution of each biometric feature group — the additional predictive value a group provides beyond what the preceding groups already capture — using HistGradientBoosting as the sole model. Quantifying marginal contribution is important because individual feature importance (as measured by SHAP) does not reveal whether an entire measurement modality is necessary; a group of features may individually show non-zero importance while collectively providing no improvement over a simpler feature set. Feature groups are evaluated cumulatively:

- Group 1 (63 features): anthropometric and metabolic features only.
- Group 2 (73 features): Group 1 plus ten bioimpedance measurements.
- Group 3 (83 features): Group 2 plus ten postural risk scores.
- Group 4 (88 features): Group 3 plus five joint angle measurements.

The same 30-split protocol, imputation procedure, and scaling procedure apply.

5.8 Statistical Testing

Pairwise statistical significance is assessed for each experiment by comparing the 30 per-fold MAE scores produced by two conditions (a pair being, for example, two models in Experiment 1, or two training regimes in Experiment 2) using the Wilcoxon signed-rank test. The null hypothesis in each test is H_0: the median difference in MAE between the two conditions is zero; the alternative hypothesis is H_1: a systematic difference in median MAE exists between the two conditions.

The Wilcoxon signed-rank test is chosen over a parametric alternative such as a paired t-test or repeated-measures ANOVA because the per-fold MAE distributions cannot be assumed normally distributed. A Shapiro–Wilk normality test applied to the 30 per-fold MAE scores confirmed non-normality in 13 of 24 tested distributions across all three experiments ($p < 0.05$). In Experiment 1, four of the five models are non-normal: GradientBoosting ($W = 0.879$, $p = 0.003$), HistGradientBoosting ($W = 0.848$, $p = 0.001$), RandomForest ($W = 0.889$, $p = 0.005$), and Ridge ($W = 0.756$, $p < 0.001$); only SVR cannot reject normality ($W = 0.964$, $p = 0.393$). In Experiment 3, all four feature-group distributions are non-normal ($p \leq 0.002$). Since parametric paired tests require normality of the difference scores for all conditions in a comparison, and the majority of distributions violate this assumption, the Wilcoxon signed-rank test is the appropriate choice throughout. The Wilcoxon signed-rank test is appropriate for paired, non-normally distributed observations and makes no assumption about the shape of the underlying distribution beyond symmetry of the difference scores.

In Experiment 1, the $\binom{5}{2} = 10$ model pairs yield a Bonferroni-corrected significance threshold of $\alpha = 0.005$. Bonferroni correction is applied to control the family-wise error rate: the probability of making at least one false positive claim of significance across all simultaneous comparisons. In Experiment 2, three regime pairs per model yield a corrected threshold of $\alpha/3 \approx 0.0167$. In Experiment 3, six group pairs yield a corrected threshold of $\alpha/6 \approx 0.0083$. A result that does not reach the corrected significance threshold is characterised as inconclusive rather than as evidence of equivalence. With 30 paired observations and a corrected threshold well below the nominal 0.05 level, the tests have limited statistical power to detect moderate effect sizes.

```
from scipy.stats import wilcoxon

n_comparisons = 10  # C(5,2) model pairs, Experiment 1
stat, p_raw = wilcoxon(mae_folds_a, mae_folds_b)
p_adjusted  = min(p_raw * n_comparisons, 1.0)
```

Listing 5.4: Paired Wilcoxon signed-rank test with Bonferroni correction.

SHAP TreeExplainer is applied to the best-performing model from Experiment 1 as a supplementary interpretability step, attributing each prediction to individual features.

```
import shap

explainer   = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X)
shap.summary_plot(shap_values, X, plot_type='bar')
```

Listing 5.5: SHAP TreeExplainer for per-prediction feature attribution.

5.9 Known Limitations of the Dataset and Feature Set

5.9.1 Sentinel Values Encoded as Numeric Features

Two features use the integer -1 as a sentinel value — a reserved numeric code used to indicate that a measurement was not computed or is not applicable for a given scan, distinct from a meaningful numeric measurement — meaning that the metric was not computed for a given scan: `sportSafeRisk` is -1 in 37% of records (approximately 58 of 158), and `sportLevel` is -1 in 70% of records (approximately 111 of 158). Both features are included in the model feature set and are passed to all models as continuous numeric inputs. The value -1 carries no ordinal meaning in this context; it is a missing-data indicator that happens to fall below the minimum meaningful score of zero. These features were retained without re-encoding because altering them would create a mismatch with the saved production model artifact that the Streamlit dashboard depends on. Proper encoding (replacing -1 with `NaN` followed by median imputation, or introducing a binary missingness indicator) is listed in future work. The presence of sentinel values may introduce a systematic artefact into the learned feature space for tree-based models and a sign-reversal artefact for linear models that treat -1 as a numerically meaningful input.

5.9.2 Near-Constant Features After Median Imputation

Five features have missingness rates exceeding 88% of records: `leftVision` (94.3%), `rightVision` (92.4%), `bloodMaxPressure` (91.8%), `bloodMinPressure` (91.8%), and `restingHeartRate` (88.6%). After training-set median imputation, these features take a single constant value for the majority of records in any given training fold, contributing negligible variance and therefore negligible predictive signal. SHAP analysis on the full production model confirms near-zero contribution from each of these features. They are retained for consistency with the production model artifact. Their presence inflates the nominal feature count without proportional contribution to predictive performance, and this should be borne in mind when interpreting the feature group ablation results in Experiment 3: the anthropometric and metabolic group nominally contains 63 features, but five of these carry essentially no independent information after imputation.

5.9.3 Redundant Derived Feature Within the Postural Risk Group

The feature `aggregated_postural_index` is the arithmetic mean of six postural risk scores (`frontHeadRisk`, `humpbackRisk`, `kneeRisk`, `pelvisRisk`, `postureRisk`, and `spineRisk`) that are themselves present as individual variables in the same feature group. The aggregate therefore carries no independent information relative to its components when all six are simultaneously available, as is the case in all experiments. The feature was included in the original model training and is retained here for consistency. The redundancy does not affect the qualitative conclusions of the feature group ablation because the postural risk group as a whole is assessed against the other groups, but it does mean that the postural risk group nominally contains ten features while functionally contributing independent information from at most nine.

5.9.4 Statistical Power of the Wilcoxon Test Under Bonferroni Correction

At 30 paired observations and a corrected α of 0.005, the Wilcoxon signed-rank test has limited power to detect moderate effect sizes in the range of Cohen's $d = 0.3$ to 0.5. The root cause is the dataset size: 53 unique individuals is insufficient for high-power pairwise testing after Bonferroni correction.

5.9.5 Unknown Client Records

Of the 158 records in the dataset, 103 carry the name value `Unknown`, representing 65% of the dataset. These records cannot be grouped by individual identity with certainty under the GroupShuffleSplit protocol. When the Unknown group falls into a test fold, all 103 of its records are evaluated by a model trained exclusively on named individuals, producing a distribution shift that drives elevated errors in those folds. This effect is discussed in Section 6.2.3 in the context of the per-prediction error distribution, and is listed here as a named dataset limitation.

5.10 Answering the Research Questions

This section maps each research sub-question to the experiment that addresses it and explains why the experimental design is the appropriate method for answering that question.

The first sub-question (which algorithm best approximates the Anovator age gap and whether any performance difference is statistically certifiable) requires an empirical comparison in which all candidate algorithms are evaluated under identical conditions on the same data. Experiment 1 provides this comparison by running all five algorithms on the same 30 cross-validation splits with the same training and test data at each fold. The pairwise Wilcoxon signed-rank test on per-fold MAE scores is the appropriate statistical instrument because the 30 fold scores are paired (drawn from the same folds) and cannot be assumed Gaussian. The Bonferroni correction controls the family-wise error rate across the $\binom{5}{2} = 10$ model pairs. The design answers the sub-question in two parts: a ranking of models by mean MAE, which identifies the best-performing algorithm by heuristic ordering, and a significance determination, which establishes whether that ranking is statistically certifiable given the available data. The design also allows a null finding (that no pairwise difference is statistically significant) to be reported as a substantive result rather than a failure, because the statistical power available at 30 observations under Bonferroni correction is explicitly characterised in Section 5.9.4.

The second sub-question (whether GMM synthetic augmentation improves approximation quality and for which model classes) requires an experiment that can attribute observed performance differences to training data composition rather than to other sources of variation. Experiment 2 achieves this attribution through the three-regime design: RealOnly, RealPlusSynth, and SynthOnly. By holding the model architecture, evaluation protocol, and test set constant across regimes while varying only the training data composition, any performance difference between regimes can be attributed specifically to the addition or substitution of synthetic records. The comparison of RealPlusSynth against RealOnly isolates the marginal benefit of synthetic augmentation given access to real data; the comparison of RealPlusSynth against SynthOnly tests whether the real records provide value beyond what the synthetic corpus alone covers. Evaluating all five models under the same regime structure further allows the sub-question's model-class specificity to be answered: whether the synthetic augmentation effect is uniform across algorithm families or depends on model architecture.

The third sub-question (which biometric feature groups carry independent predictive information relative to the Anovator score) requires an experiment that can isolate the marginal contribution of each group while holding the preceding groups constant. Experiment 3 achieves this through cumulative additive evaluation: Group 1 is assessed alone, Group 2 as Group 1 plus bioimpedance features, Group 3 as Group 2 plus postural risk features, and Group 4 as Group 3 plus joint angle features. The sequential structure ensures that any observed performance change at each step can be attributed specifically to the features added at that step, not to changes in the base feature set. Using a single model (HistGradientBoosting, selected by its ranking in Experiment 1) controls for model-level variance, ensuring that observed differences across feature groups are not confounded with differences in how distinct algorithm families utilise the same feature set. The Wilcoxon tests on the $\binom{4}{2} = 6$ group pairs, with Bonferroni correction, provide the formal statistical determination of whether any group transition produces a certifiable change in prediction quality at the available sample size.

Chapter 6

Results and Conclusions

6.1 Preliminary Note on R^2

All three experiments report mean R^2 alongside MAE and RMSE for completeness. However, R^2 must be interpreted with caution for two reasons. First, each test fold contains between 11 and 13 records, and with test sets of this size R^2 estimates are extremely noisy. Second, and more importantly, all five models return negative mean R^2 values under the RealOnly regime, which means their predictions are less accurate than simply predicting the mean of the test fold for every observation. This is not unexpected given the combination of small test folds, high outcome variance (age_gap standard deviation of 3.30 years across 158 records), and the GroupShuffleSplit structure which occasionally assigns the 103 Unknown records entirely to the test fold, producing a sharp distribution shift. Negative R^2 values in this context reflect the difficulty of the evaluation protocol rather than a fundamental failure of the models; MAE and RMSE on the same folds remain in the 2.4–3.4 year range and are the appropriate metrics for assessing approximation quality.

6.2 Experiment 1: Model Comparison

6.2.1 Performance Across Metrics

Table 6.1 presents mean MAE, RMSE, and R^2 with 95% confidence intervals for all five models, derived from 30 GroupShuffleSplit folds.

Table 6.1: Experiment 1: Mean performance metrics (95% CI) across 30 folds. Models ordered by ascending mean MAE.

Model	MAE mean	MAE 95% CI	RMSE mean	RMSE 95% CI	R^2 mean	R^2 95% CI
HistGradientBoosting	2.43	[2.09, 2.76]	3.05	[2.66, 3.44]	−0.36	[−0.70, −0.03]
Ridge	2.48	[2.09, 2.87]	3.10	[2.62, 3.58]	−0.49	[−1.06, 0.08]
RandomForest	2.51	[2.18, 2.84]	3.20	[2.85, 3.56]	−0.47	[−0.77, −0.17]
GradientBoosting	2.51	[2.19, 2.84]	3.13	[2.75, 3.51]	−0.42	[−0.74, −0.10]
SVR	2.62	[2.30, 2.94]	3.42	[3.11, 3.73]	−0.66	[−0.91, −0.41]

Note: R^2 values are reported for completeness only. Test folds contain 11–13 records; at this scale R^2 estimates are highly variable and negative values are expected. MAE and RMSE are the primary metrics throughout.

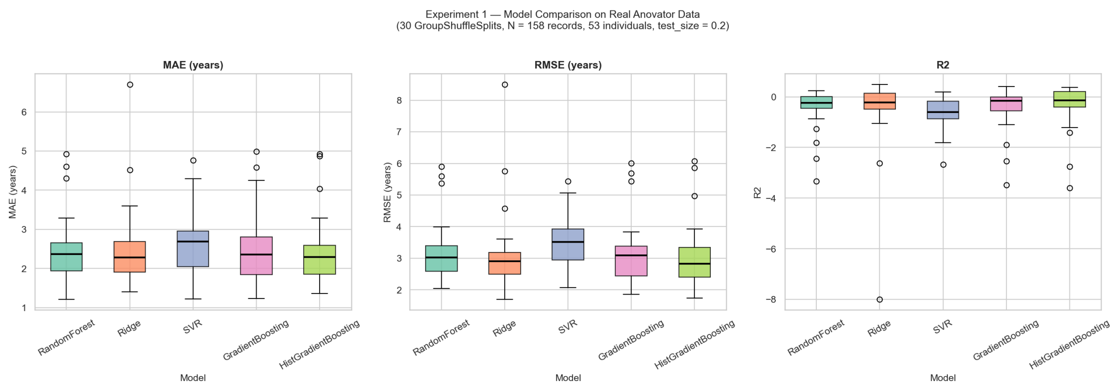

Figure 6.1: Box plot of MAE distributions across 30 cross-validation folds for all five models (Experiment 1). Models are ordered by ascending median MAE. The horizontal line within each box represents the median; the box spans the interquartile range; whiskers extend to 1.5 times the IQR.

HistGradientBoosting achieves the lowest mean MAE at 2.43 years and the lowest mean RMSE at 3.05 years. SVR has the highest mean MAE at 2.62 years. The 95% confidence intervals for MAE overlap extensively across all five models, consistent with the significance test results in Section 6.2.2.

6.2.2 Statistical Significance

Table 6.2 presents the Wilcoxon signed-rank test results for all ten model pairs on MAE, with Bonferroni-corrected p-values and the corrected significance threshold of 0.005.

Table 6.2: Experiment 1: Wilcoxon signed-rank test results for MAE (Bonferroni $\alpha = 0.005$).

Model A	**Model B**	p **(raw)**	p **(adjusted)**	**Significant**
SVR	HistGradientBoosting	0.0043	0.0434	No
RandomForest	HistGradientBoosting	0.0577	0.5769	No
RandomForest	SVR	0.0636	0.6356	No
Ridge	SVR	0.0767	0.7672	No
GradientBoosting	HistGradientBoosting	0.1294	1.0000	No
RandomForest	Ridge	0.2710	1.0000	No
Ridge	GradientBoosting	0.2710	1.0000	No
SVR	GradientBoosting	0.3599	1.0000	No
RandomForest	GradientBoosting	0.8236	1.0000	No
Ridge	HistGradientBoosting	0.9515	1.0000	No

None of the ten pairwise comparisons reaches the Bonferroni-corrected threshold of 0.005. All ten results must be characterised as inconclusive: the available data cannot establish whether any two models differ in their ability to approximate the Anovator bodyAge score at the required level of statistical confidence.

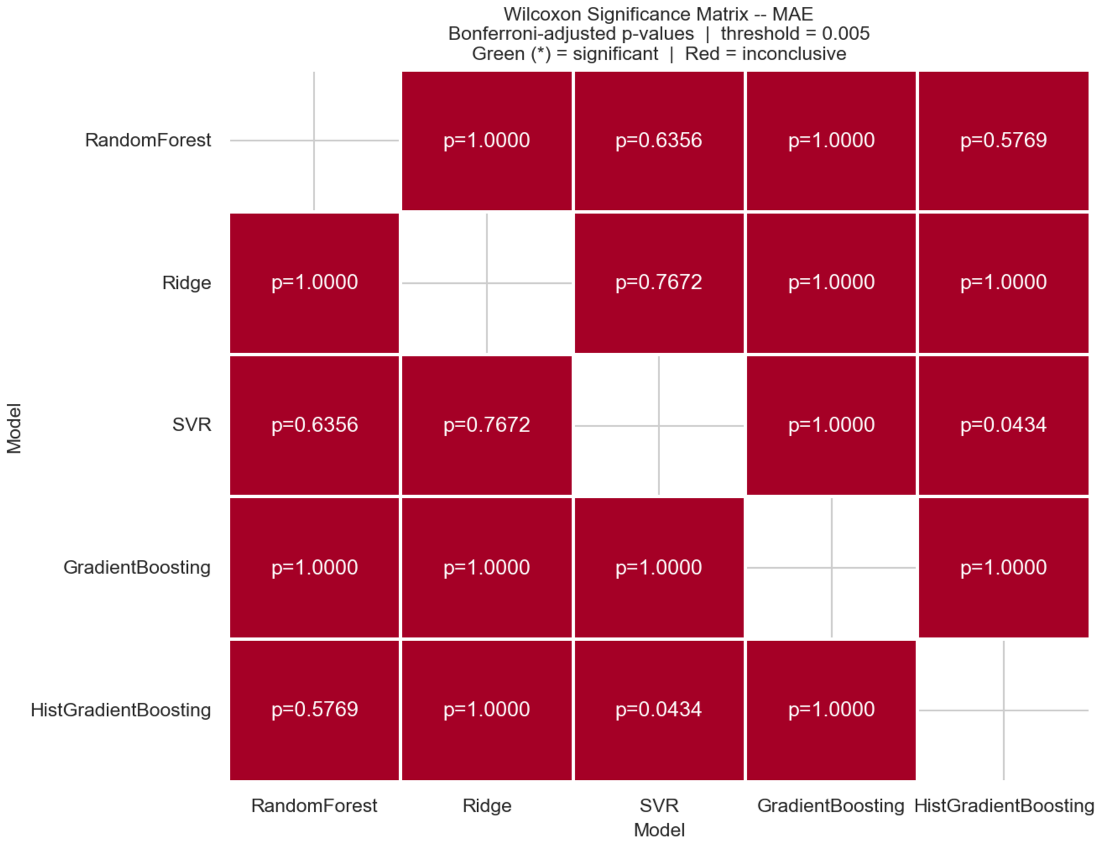

Figure 6.2: Pairwise significance matrix for Experiment 1 model comparisons. All ten comparisons are inconclusive at the Bonferroni-corrected threshold of $\alpha = 0.005$.

6.2.3 Per-Prediction Error Distribution and Clinical Reliability

Table 6.3 presents the per-prediction absolute error distribution for HistGradientBoosting across all 655 test predictions in the 30 folds.

Table 6.3: Per-prediction absolute error distribution (HistGradientBoosting, 655 test predictions across 30 folds).

Tolerance	Predictions within tolerance	Percentage
$\leq$ 0.5 years	77 / 655	11.8%
$\leq$ 1.0 years	144 / 655	22.0%
$\leq$ 2.0 years	247 / 655	37.7%
$\leq$ 3.0 years	326 / 655	49.8%

The median absolute error across 655 individual predictions is 3.00 years; the 90th percentile is 6.84 years. These figures are substantially higher than the 2.43-year mean fold-level MAE reported in Table 6.1. The discrepancy arises from the GroupShuffleSplit structure interacting with the 103 records labelled “Unknown” in the name field (see Section 5.9.5). When the Unknown group falls into a test fold, all 103 of its records are evaluated by a model trained exclusively on named individuals, producing a sharp distribution shift that drives high errors in those folds.

6.2.4 Conclusion: Sub-question 1

HistGradientBoosting is the numerically preferred model with a mean MAE of 2.43 years, but no algorithm can be certified as statistically superior to any other: all ten Bonferroni-corrected Wilcoxon tests are inconclusive at the corrected threshold of $\alpha = 0.005$. The 53-individual dataset does not provide sufficient statistical power to resolve the performance differences observed, and Sub-question 1 can be answered in two parts: HistGradientBoosting is the numerically preferred model by heuristic ranking, but no algorithm can be certified as statistically superior to the others at the available sample size.

6.3 Experiment 2: Synthetic Data Ablation

6.3.1 Synthetic Data Fidelity

The quality of the GMM-generated synthetic population was assessed by comparing its pairwise correlation structure to that of the 158 real training records. The correlation matrix was computed over all 79 shared numeric features. The mean absolute pairwise correlation difference between the real and synthetic matrices is 0.0244, and the Frobenius norm of the difference matrix is 2.46 over a 79×79 matrix (equivalent to a normalised value of 0.031 per feature). These figures indicate that the GMM synthesis preserves inter-feature relationships closely.

6.3.2 Performance by Regime and Model

Table 6.4 presents mean MAE with 95% confidence intervals for all five models under each of the three training regimes.

Table 6.4: Experiment 2: Mean MAE (95% CI) by model and training regime. All test evaluations use the same real held-out folds with true age_gap labels.

Model	**RealOnly**	**RealPlusSynth**	**SynthOnly**
HistGradientBoosting	2.43 [2.09, 2.76]	2.09 [1.85, 2.32]	2.07 [1.84, 2.30]
GradientBoosting	2.51 [2.19, 2.84]	2.15 [1.93, 2.37]	2.07 [1.85, 2.30]
RandomForest	2.51 [2.18, 2.84]	2.26 [2.05, 2.47]	2.17 [1.93, 2.40]
Ridge	2.48 [2.09, 2.87]	2.39 [2.15, 2.63]	3.23 [2.14, 4.31]
SVR	2.62 [2.30, 2.94]	2.28 [2.06, 2.49]	2.24 [2.02, 2.45]

Under the RealPlusSynth regime, four of the five models show a reduction in mean MAE relative to RealOnly, with improvements ranging from 0.09 years for Ridge to 0.36 years for HistGradientBoosting. Ridge degrades substantially under the SynthOnly regime to a mean MAE of 3.23 years, the single largest deterioration observed across the entire experiment. The SynthOnly Ridge fold distribution is also strikingly non-normal (Shapiro–Wilk $W = 0.538$, $p < 0.001$), reflecting the instability of a linear model trained on pseudo-labels with artificially compressed variance; the fold scores are highly skewed rather than centred around a stable mean.

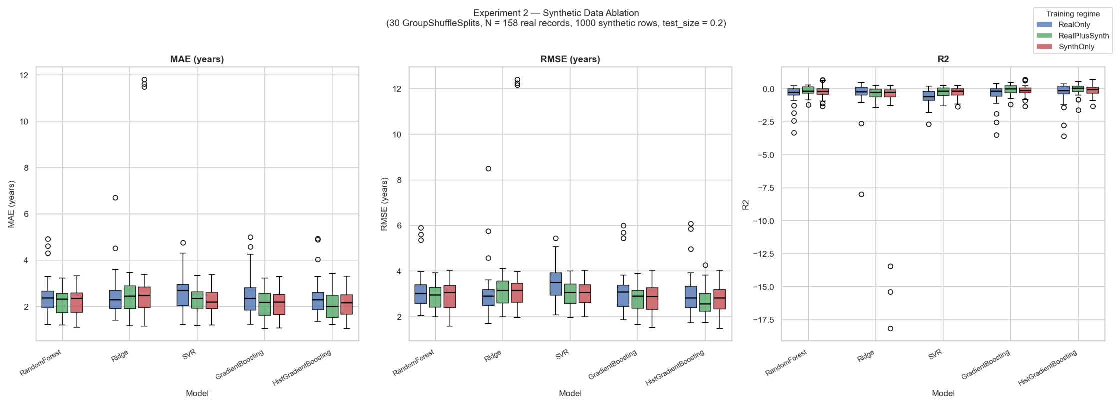

Figure 6.3: Box plot of MAE distributions by training regime and model (Experiment 2). The Ridge SynthOnly distribution has a substantially wider spread and higher median than all other conditions.

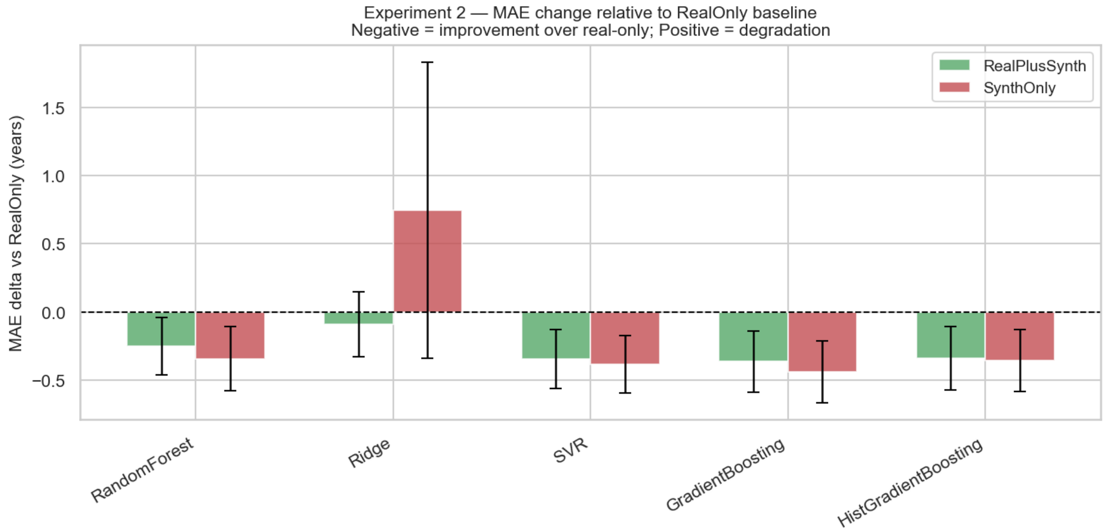

Figure 6.4: Mean MAE delta relative to the RealOnly baseline for each model under RealPlusSynth and SynthOnly training regimes (Experiment 2). Negative values indicate improvement over the RealOnly baseline.

6.3.3 Statistical Significance

Table 6.5 presents the statistically significant pairwise comparisons from Experiment 2 on MAE. Bonferroni correction is applied over three regime pairs per model, yielding a corrected threshold of 0.0167.

Five comparisons reach significance. For HistGradientBoosting, RandomForest, GradientBoosting, and SVR, the RealOnly versus RealPlusSynth comparison is statistically significant, with adjusted p-values ranging from 0.000004 to 0.002855. No regime pair comparison reaches significance for Ridge. The RealPlusSynth versus SynthOnly comparison is not significant for any model.

Table 6.5: Experiment 2: Significant Wilcoxon results for MAE (Bonferroni $\alpha = 0.0167$). Only comparisons reaching significance are shown; all remaining pairs are inconclusive.

Model	**Regime A**	**Regime B**	p **(raw)**	p **(adjusted)**
HistGradientBoosting	RealOnly	RealPlusSynth	< 0.000001	0.000004
SVR	RealOnly	RealPlusSynth	0.000003	0.000010
SVR	RealOnly	SynthOnly	0.000027	0.000081
RandomForest	RealOnly	RealPlusSynth	0.000111	0.000332
GradientBoosting	RealOnly	RealPlusSynth	0.000952	0.002855

6.3.4 Conclusion: Sub-question 2

GMM synthetic augmentation significantly improves approximation quality for four of the five models under the RealPlusSynth regime, with adjusted p-values ranging from 0.000004 to 0.002855 and mean MAE reductions of 0.09 to 0.36 years. Ridge is the exception: training on pseudo-labelled synthetic data alone degrades its MAE by 30%, because the compressed pseudo-label variance (standard deviation 1.91 versus 3.30 years in real labels) biases the linear fit toward a narrower output range than the true target distribution.

6.4 Experiment 3: Feature Group Ablation

6.4.1 Performance by Feature Group

Table 6.6 presents mean MAE, RMSE, and R^2 with 95% confidence intervals for HistGradientBoosting under each of the four cumulative feature group conditions.

Table 6.6: Experiment 3: Mean performance metrics (95% CI) by feature group (HistGradientBoosting).

Group	**Feat.**	**MAE**	**MAE CI**	**RMSE**	**RMSE CI**	R^2	R^2 **CI**
G1: Anthropometric + Metabolic	63	2.39	[2.05, 2.72]	3.01	[2.62, 3.40]	−0.33	[−0.66, −0.01]
G2: + Bioimpedance	73	2.42	[2.07, 2.76]	3.05	[2.65, 3.46]	−0.37	[−0.72, −0.03]
G3: + Postural Risk	83	2.42	[2.08, 2.76]	3.06	[2.67, 3.46]	−0.37	[−0.71, −0.04]
G4: + Joint Angles (full)	88	2.43	[2.09, 2.76]	3.05	[2.66, 3.44]	−0.36	[−0.70, −0.03]

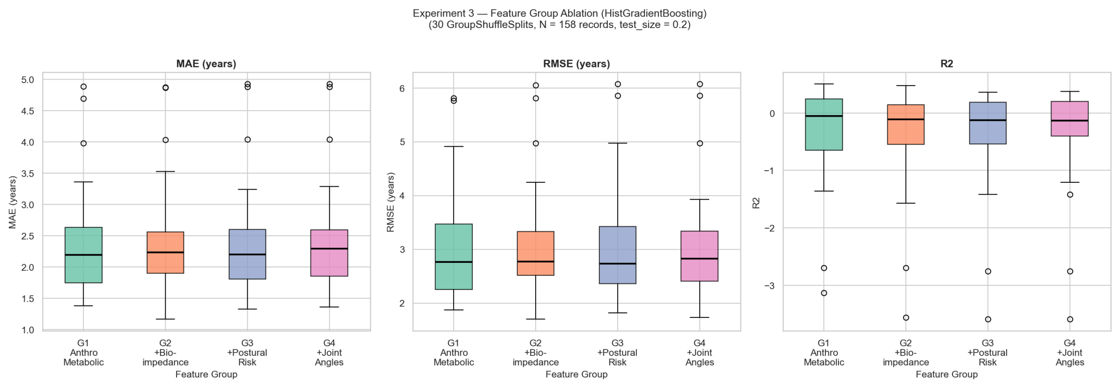

Figure 6.5: Box plot of MAE distributions by cumulative feature group (Experiment 3, HistGradientBoosting).

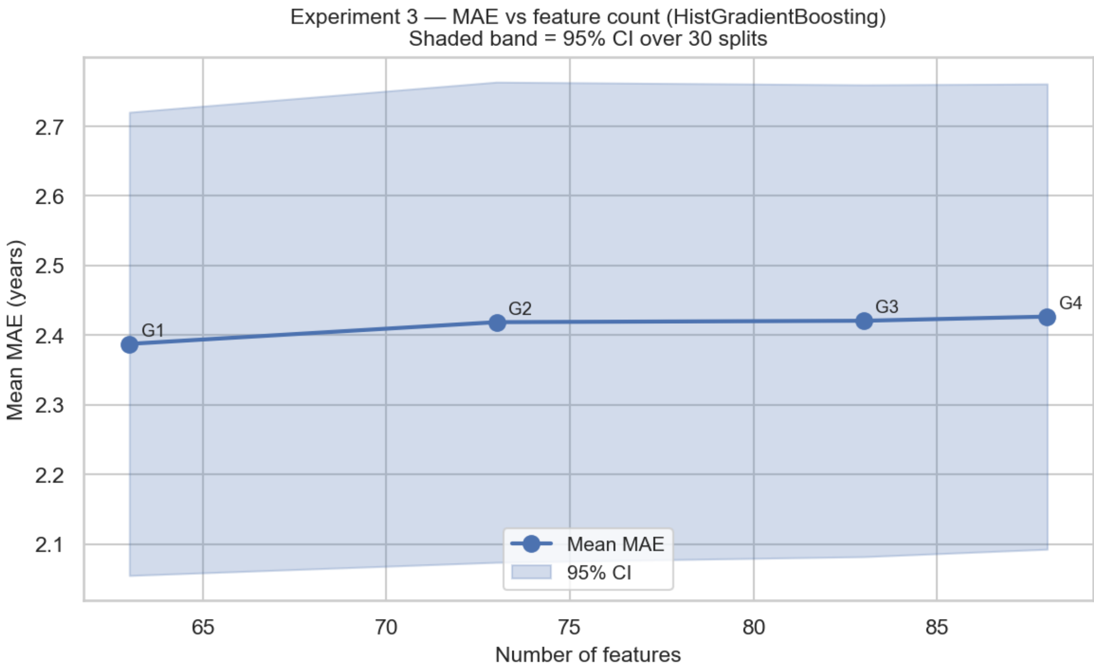

Figure 6.6: Mean MAE by cumulative feature group (Experiment 3). Error bars represent 95% confidence intervals across 30 cross-validation folds. The monotonic increase from G1 to G4 is not statistically significant.

The anthropometric and metabolic group (G1) achieves the lowest mean MAE at 2.39 years. Each successive addition of a feature group is associated with a marginal increase in mean MAE. The total difference between the 63-feature group and the full 88-feature set is 0.04 years.

6.4.2 Statistical Significance

Table 6.7 presents the Wilcoxon signed-rank test results for all six group pairs on MAE, with Bonferroni-corrected p-values and the corrected significance threshold of 0.0083.

Table 6.7: Experiment 3: Wilcoxon signed-rank test results for MAE (Bonferroni $\alpha = 0.0083$).

Group A	**Group B**	p **(raw)**	p **(adjusted)**	**Sig.**
G1: Anthropometric + Metabolic	G3: + Postural Risk	0.2207	1.0000	No
G1: Anthropometric + Metabolic	G4: + Joint Angles	0.2129	1.0000	No
G1: Anthropometric + Metabolic	G2: + Bioimpedance	0.2710	1.0000	No
G2: + Bioimpedance	G4: + Joint Angles	0.4280	1.0000	No
G3: + Postural Risk	G4: + Joint Angles	0.5481	1.0000	No
G2: + Bioimpedance	G3: + Postural Risk	0.5699	1.0000	No

None of the six pairwise comparisons reaches the Bonferroni-corrected threshold of 0.0083. All six comparisons are therefore inconclusive.

6.4.3 Conclusion: Sub-question 3

No biometric feature group beyond the 63-feature anthropometric and metabolic base achieves a statistically significant improvement at the Bonferroni-corrected threshold of $\alpha = 0.0083$. All six pairwise feature group comparisons are inconclusive, and the base group alone achieves the lowest mean MAE at 2.39 years, suggesting that bioimpedance, postural risk, and joint angle measurements carry negligible independent weight in the Anovator formula at this data scale.

6.5 Answers to the Research Sub-Questions

The first sub-question asked which regression algorithm best approximates the Anovator age gap and whether any performance difference is statistically certifiable. The answer is two-part: HistGradientBoosting is the numerically preferred model by heuristic ranking, and no algorithm can be certified as statistically superior to the others at the available sample size.

The second sub-question asked whether GMM synthetic augmentation improves approximation quality and for which model classes. For ensemble methods and SVR the answer is yes, with statistically significant improvements under the RealPlusSynth regime. For Ridge the answer is no: synthetic augmentation through teacher-student pseudo-labelling is contraindicated when pseudo-label variance is substantially compressed relative to the real target distribution.

The third sub-question asked which biometric feature groups carry independent predictive information. No statistical evidence exists for independent predictive contribution from bioimpedance, postural risk, or joint angle features beyond the anthropometric and metabolic base, but equivalence cannot be asserted either; all comparisons are inconclusive at the available sample size.

6.6 Limitations

Four dataset and methodological limitations are documented in full in Section 5.9. In brief: two features use the integer -1 as a sentinel for missing data and are treated as continuous numeric inputs; five features with missingness above 88% become near-constants after imputation; the derived feature `aggregated_postural_index` is arithmetically redundant within the postural risk group; and the Wilcoxon signed-rank test at Bonferroni-corrected thresholds has limited power with 30 paired observations. The 53-individual, single-centre dataset is the binding constraint throughout.

Chapter 7

Discussion and Recommendations

7.1 The Null Result in Experiment 1 Is a Finding, Not a Failure

The absence of statistical significance across all ten pairwise model comparisons in Experiment 1 is the most important single result in this thesis and must be stated precisely. The Wilcoxon signed-rank test on 30 paired fold observations, corrected for ten simultaneous comparisons at $\alpha = 0.005$, found no evidence that any two of the five models differ in their ability to approximate the Anovator bodyAge score. The adjusted p-value for the closest comparison (SVR versus HistGradientBoosting) was 0.043, an order of magnitude above the corrected threshold.

Non-significance does not mean the models are equivalent. Claiming equivalence from a non-significant Wilcoxon test would commit the fundamental error of accepting the null hypothesis. What the result means is that the study, as constituted, cannot resolve the performance difference between these models with the required statistical confidence. The confidence intervals reported in Table 6.1 span approximately 0.65 to 0.78 years, a range that is large relative to the mean differences between models, which are all below 0.20 years.

The finding has scientific content precisely because it is unexpected. The fact that Ridge regression achieves mean MAE within 0.05 years of HistGradientBoosting suggests either that the predictive information in these 88 features is largely captured by linear combinations of those features, implying that Anovator's proprietary formula may itself be approximately linear in the inputs; or that the test folds are too small (averaging 12 records drawn from 11 individuals) to discriminate non-linear signal from noise in a dataset of this size. Both interpretations are scientifically meaningful and are discussed further in the context of the feature group results below.

The practical implication is equally clear. For any operational deployment of a model intended to approximate the Anovator score within this data context, the choice among these five algorithms does not matter in a statistically defensible sense. The selection of HistGradientBoosting as the model for Experiment 3 is justified by its numerical rank at the top of the MAE ordering, but this selection carries an explicit caveat that it represents a heuristic choice rather than a certified one.

7.2 Synthetic Augmentation Helps Ensemble Methods and Damages Ridge

Experiment 2 produces the strongest statistical results in the study. The mechanism is interpretable within the teacher-student framework. Tree-based ensemble methods and kernel SVMs benefit from larger training sets because they rely on local density estimation or partitioning that becomes more reliable as the coverage of the input space increases. The 1000 synthetic records fill gaps in the biometric input space that the 53 individuals in the real dataset leave sparsely covered.

Ridge regression is the exception. Under the SynthOnly regime, Ridge's mean MAE rises to 3.23 years from a RealOnly baseline of 2.48 years (a 30% degradation), and the associated confidence interval widens to [2.14, 4.31]. The degradation follows directly from the pseudo-label distribution: the production model's predictions on the synthetic records have a standard deviation of 1.91 years compared to 3.30 years in the real label distribution. Linear models fit their coefficients to minimise squared error across the training distribution; when the training distribution artificially compresses the label variance, the fitted linear model learns a relationship calibrated to a narrower output range than the true target. When this model is then evaluated on real test folds where age_gap ranges more widely, the mismatch between training label variance and test label variance produces systematically biased predictions. Tree-based and kernel methods are less susceptible to this effect because they make fewer assumptions about the global shape of the target distribution and are more responsive to local input-output structure.

Synthetic augmentation through teacher-student pseudo-labelling is not model-agnostic. Before applying it in an operational context, the distributional alignment between real and pseudo-label variance should be verified, and linear models should either be excluded from synthetic training or trained on a label distribution that has been re-scaled to match the real target variance.

The RealPlusSynth versus SynthOnly comparison is not significant for any model. The non-significance does not mean the real records are redundant. Under RealOnly, all five models achieve lower MAE than the pseudo-labels' intrinsic noise floor suggests should be achievable from teacher predictions alone, which implies that the true labels in the 158 records carry information the teacher has not fully extracted. Rather, the failure to detect a significant difference between RealPlusSynth and SynthOnly reflects the limited power available to detect what is likely a small marginal contribution of approximately 100 to 130 real records against a 1000-record synthetic corpus.

7.3 Feature Group Saturation and What It Implies About Anovator's Formula

The most counterintuitive finding in the study is that the 63-feature anthropometric and metabolic group alone achieves lower mean MAE (2.39 years) than any larger feature set, including the full 88-feature model (2.43 years). Several interpretations are consistent with this result. The first, and most important for the thesis framing, concerns the structure of the Anovator formula itself. Anovator's bodyAge computation is proprietary and unpublished. If the formula places low or negligible weight on bioimpedance raw values, postural camera risk scores, and joint angles in favour of anthropometric and metabolic measurements, then a model attempting to reverse-engineer the formula would find no additional predictive information in those feature groups beyond what the base group already captures.

The second interpretation concerns the training set size. With approximately 145 real records per training fold, adding 10, 20, or 25 features to a 63-feature model introduces a modest increase in the curse of dimensionality that a tree-based algorithm must navigate. HistGradientBoosting's gradient boosting procedure provides some protection against overfitting, but the added feature dimensions still increase the number of candidate splits evaluated at each node and may dilute signal with noise in biometric measurements that show limited variance across the 53 individuals in the dataset.

The third interpretation concerns data quality in the added feature groups. The bioimpedance values (`r20` and `r100` measurements) depend on electrode placement and individual hydration status at the time of scanning, introducing measurement noise. The postural risk scores are categorical or ordinal outputs of a computer vision algorithm applied to static posture images and may be less reliable across sessions for the same individual. The joint angles are similarly image-derived estimates. If these measurements carry meaningful signal for the Anovator score, that signal may be buried under measurement noise at the scale of 158 records.

Whichever interpretation is correct, the operational implication is consistent: at this data scale, there is no statistical justification for preferring the full 88-feature model over the 63-feature anthropometric and metabolic subset. A repeat of this experiment on a dataset of several hundred individuals would carry sufficient power to detect whether the additional feature groups contribute marginal predictive value. Until then, the finding must be reported as inconclusive on the direction of the true marginal effect, and as consistent with the hypothesis that Anovator's scoring places primary weight on anthropometric and metabolic variables.

7.4 SHAP Feature Importance: Physiological Interpretation

Global SHAP values reveal a pronounced concentration of predictive weight in anthropometric and metabolic features. Waist circumference (`wc`) is the single dominant predictor with a mean absolute SHAP value of 0.69 years, more than twice that of any other feature. Its dominance is consistent with its likely role as a primary input to the Anovator bodyAge formula: central obesity is the biometric signal most directly linked to the cardiometabolic outcomes that commercial biological age scores typically aim to capture.

The second and third most influential features, `aerobicGoal` and `agility`, warrant a caveat. `AerobicGoal` is an Anovator-computed training recommendation, not a raw biometric measurement, meaning the surrogate model has partially learned to use Anovator's own intermediate outputs as predictors of the final score, which is expected in a reverse-engineering setting but limits physiological interpretability. Intra-abdominal fat (`inFat`) and BMI follow as the fourth and sixth most important features respectively. The first bioimpedance feature (`r20LeftLeg`) appears at rank five, and the first joint angle feature appears at rank eleven, well below the anthropometric and metabolic cluster. This gradient directly supports the interpretation that Anovator's formula draws its predictive structure primarily from body composition, fitness, and metabolic measurements.

7.5 The Underpowered Wilcoxon Caveat and Its Scope

The statistical power limitation documented in Section 5.9.4 applies differently across the three experiments. In Experiment 1 it is binding: all ten adjusted p-values exceed the corrected threshold of 0.005, and the 0.19-year mean MAE difference between the closest pair (SVR versus HistGradientBoosting) implies a modest effect size that may remain undetectable even with 50 folds. In Experiment 2 it is not binding for the five significant comparisons, which reach adjusted p-values as low as 0.000004; it is relevant only for the non-significant RealPlusSynth-versus-SynthOnly comparisons. In Experiment 3 it is clearly binding: the lowest raw p-value is 0.21, a value that would not reach even the uncorrected 0.05 threshold.

Non-significant results from an underpowered test are not failures. They reflect the honest limits of what 53 individuals and 30 cross-validation folds can resolve at Bonferroni-corrected thresholds, and must be reported as such.

7.6 Known Limitations and Their Bearing on Interpretation

The four limitations documented in Section 5.9 bear on the interpretation of results as follows. The sentinel encoding of `sportSafeRisk` and `sportLevel` may introduce systematic artefacts into the learned feature space but does not affect the comparability of results across experiments, since all conditions share the same encoding. The near-constant imputed features inflate the nominal dimensionality without proportional contribution to predictive performance. The redundant `aggregated_postural_index` has negligible practical effect on tree-based models due to their tolerance for collinear features. Finally, the single-centre, 53-individual dataset means that all findings must be characterised as an analysis of the reverse-engineering problem within this specific dataset rather than a generalisation claim at population scale.

7.7 Synthesis

Taken together, the three experiments produce a coherent picture. The five models are statistically indistinguishable on the real-data-only approximation task, suggesting that the predictive information available in the 88 features is largely captured by their linear structure and by the anthropometric and metabolic subset in particular. Synthetic augmentation through teacher-student pseudo-labelling produces statistically significant improvements for ensemble methods and SVR, indicating that the 53-individual training corpus is the primary constraint on approximation quality. Ridge's failure under the SynthOnly regime provides a clear case study in the conditions under which synthetic augmentation is contraindicated.

7.8 Recommendations for Zdravoletie

The following recommendations are drawn from the methodological limitations and open questions identified in this study. Each is framed as a concrete next step that Zdravoletie can take to extend and operationalise the work.

Expand the dataset across multiple clinics. The 53-individual, single-centre corpus is the binding constraint on statistical power throughout all three experiments. All ten model comparisons in Experiment 1 and all six feature group comparisons in Experiment 3 are inconclusive at the Bonferroni-corrected thresholds, not because the performance differences are zero, but because the available sample provides insufficient resolution to certify them. Coordinating data collection with other Anovator-equipped centres would reduce single-clinic bias by introducing demographic and lifestyle diversity that the Zdravoletie cohort alone cannot provide. The single-centre limitation is documented in Section 7.6: all findings in this study must be characterised as an analysis of the reverse-engineering problem within this specific dataset rather than a generalisation claim at population scale, precisely because the 53 individuals at Zdravoletie represent a narrow demographic and lifestyle range.

A dataset of several hundred individuals would raise the power of pairwise significance tests above the current inconclusive regime, support time-based evaluation splits not feasible with the current longitudinal density, and permit reliable comparison of alternative synthetic data generation architectures whose training regimes require substantially larger corpora. This claim is grounded in the power analysis documented in Section 5.9.4: at 30 paired observations and a Bonferroni-corrected threshold of $\alpha = 0.005$, the Wilcoxon signed-rank test has limited power to detect moderate effect sizes in the range of Cohen's $d = 0.3$ to 0.5. A prospective power analysis targeting 80% power at the same corrected thresholds and observed effect sizes would quantify the exact recruitment target required.

Complete GDPR requirements before production deployment. The current Streamlit application is a research prototype. Before placing it in clinical use, Zdravoletie should commission a Data Protection Impact Assessment under Article 35 GDPR, collect explicit client consent covering analytical use of scan data, execute a Data Processing Agreement with the hosting provider, and implement staff-only authentication with audit logging.

Re-encode sentinel values before any further modelling. The features `sportSafeRisk` and `sportLevel` use -1 to signal missingness and are currently passed to all models as continuous numeric inputs. Replacing -1 with `NaN` and adding a binary missingness indicator for each feature is the most contained and highest-impact preprocessing fix available; it requires no new data collection.

Evaluate alternative synthetic generation methods once the dataset exceeds 500 records. CTGAN, TVAE, and Gaussian copula approaches are excluded from the current study because they require substantially larger training corpora to train reliably. Once the real dataset reaches approximately 500 records, a direct comparison of these methods against the GMM baseline used here would be well-powered and informative.

Test imputation strategy sensitivity. The current study uses training-fold median imputation for all features with missing values. Evaluating KNN or MICE imputation as alternatives would confirm whether the key results across all three experiments are robust to this modelling choice, particularly for the five features with missingness above 88%.

Commission a formal power analysis before dataset expansion. Before investing in multi-centre data collection, a power analysis would quantify the sample size required to achieve adequate power at the Bonferroni-corrected thresholds used here for effect sizes of practical relevance, giving the expansion effort a concrete and justifiable recruitment target.

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Appendices

Appendix A

Ethical Considerations

A.1 Data Classification Under GDPR

The dataset used in this study consists of Anovator body scan records collected from clients of Zdravoletie. Under the General Data Protection Regulation (EU) 2016/679, this data falls into two protected categories. First, the name field constitutes personal data under Article 4(1), as it directly identifies natural persons. Second, the biometric and health measurement fields (including body composition percentages, impedance values, postural risk scores, blood pressure, heart rate, vision acuity, and the derived bodyAge score) constitute special category data under Article 9(1), specifically data concerning health.

Of the 158 records in the dataset, 103 carry the name value "Unknown". The remaining 50 records carry values that appear to be dates of birth encoded as six-digit strings, which are pseudonyms rather than true names but remain linkable to individuals within the clinic's operational records. The `original_url` and `id` fields contain Anovator platform identifiers that could be used to retrieve additional personal data; these fields were not used in any modelling or analysis and are excluded from the model feature set.

A.2 Consent Assumptions

The data were collected in the course of normal clinical operations at Zdravoletie. It is assumed that Zdravoletie's standard client intake process includes consent to collection and processing of biometric data for health assessment purposes. However, this study did not verify whether clients were specifically informed that their scan data would be used for machine learning research or shared with a student researcher. This represents a gap relative to the Article 9(2)(a) explicit consent standard for research use of special category data.

A.3 Data Handling in This Study

The dataset was transferred to the researcher's local development environment as a CSV export and was not uploaded to any public repository, cloud storage service, or third-party platform during analysis. All modelling, experiment scripts, and visualisations were executed locally. The Streamlit dashboard does not transmit client data to external servers; all prediction and SHAP computation runs client-side within the Streamlit session. Model artifact files (`.joblib`) stored in the repository contain only fitted model parameters and no training records.

A.4 Requirements for Production Deployment

A production deployment of the Streamlit dashboard or any successor system would require several steps before going live. A Data Protection Impact Assessment (DPIA) under Article 35 GDPR must be completed. Explicit informed consent forms covering research and analytical use of scan data must be in place, alongside a Data Processing Agreement under Article 28 GDPR with any cloud hosting provider. Access must be restricted to authorised clinic staff through authentication, with audit logging of record access. A documented data retention policy is also required for compliance with Article 17 erasure requests.

Appendix B

Use of Generative Artificial Intelligence

B.1 Declaration

Generative AI tools were used in specific, limited parts of this project. All experimental design, Python code for the three experiments, statistical analysis, results interpretation, and thesis chapter writing were completed independently by the author without AI assistance. AI tools were used in the five areas documented below, in accordance with Fontys University of Applied Sciences guidelines on the responsible use of generative AI.

B.2 Streamlit Application Development

Claude Code (Anthropic) was used to assist with development of the Streamlit web application described in Chapter 3. This included generating and debugging Python code for the application pages, implementing UI improvements, and resolving deployment issues on Streamlit Community Cloud. The author directed all functional requirements, reviewed all generated code, and made all design decisions. The tool accelerated implementation but did not contribute to the analytical or research content of the project.

B.3 LaTeX Document Preparation

Claude Code was used to convert the competency portfolio from Markdown to LaTeX and to assist with compilation and debugging of both documents. No content was generated by AI in this process; the tool was used solely for formatting, typesetting, and resolving compilation errors.

B.4 Methodology Ideation and Documentation Research

During the methodology design phase, a conversational AI assistant (Claude, Anthropic) was used to survey available synthetic data generation approaches and identify relevant implementation documentation. Approaches discussed included Gaussian Mixture Models, CTGAN, TVAE, and copula-based methods. The selection of the Gaussian Mixture Model and its full implementation in scikit-learn were carried out independently by the author based on this research. The AI served as a documentation and literature search aid rather than a code or content generator.

B.5 Editorial Review

A conversational AI assistant (Claude, Anthropic) was used to review drafts of the thesis and competency portfolio for consistency, terminology, cross-references, and phrasing. All substantive content, arguments, and conclusions were written by the author. The AI's role was equivalent to a copy-editing function: identifying inconsistencies, flagging unclear sentences, and suggesting alternative phrasing which the author then accepted, modified, or rejected independently.

B.6 Version Control and Workflow Assistance

Claude Code was used to assist with Git commit messages, push commands, and logbook formatting during the project. No analytical or research content was produced through this assistance.

B.7 Example Prompts

The following prompts are representative examples of AI tool use during the project:

```
1. "Review Section 6.2 for consistency between the table values
   and the prose descriptions. Flag any discrepancies."

2. "Fix the fl ligature encoding issue in competency_portfolio.tex
   by checking the preamble for \usepackage[T1]{fontenc}."

3. "In pages/1_Individual_Report.py, replace the plain st.metric
   calls with colour-coded styled cards that show green for
   negative age gap and amber for positive."

4. "What synthetic data generation methods are suitable for
   tabular datasets with under 200 records? Provide links to
   scikit-learn documentation for GMM implementation."

5. "Convert thesis/competency_portfolio.md to a standalone LaTeX
   document with the same preamble as main.tex."
```